UNIVERSITY OF SCIENCE AND TECHNOLOGY
OF SOUTHERN PHILIPPINES
Alubijid | Balubal |Cagayan de Oro | Claveria | Jasaan | Oroquieta | Panaon | Villanueva

Lab Activity 2

IT325 Machine Learning | Performing EDA for the PIT Dataset

Type	Laboratory Activity
Topic Focus	Exploratory data analysis for text classification datasets
Time Allotment	1 laboratory meeting
PIT Connection	Students generate EDA outputs that will support preprocessing and model selection
Mode	Individual hands on work

Learning Outcomes

1. Load and inspect a text classification dataset using Python.
2. Detects missing values, duplicates, class imbalance, and basic text characteristics.
3. Present and interpret simple EDA visualizations.

Task Overview

Students will implement the EDA plan prepared in Lecture Activity 1. The output of this task will become part of the PIT notebook and may also be used in the final presentation.

Laboratory Tasks

1. Load the dataset in pandas and display the first few rows.
2. Show the dataset shape, column names, and data types.
3. Check for missing values and duplicate rows.
4. Show the distribution of the target variable.
5. Create at least five EDA visualizations relevant to the dataset.
7. In a Text Cell Write three to five observations based on the analysis.

Minimum Required Outputs

1. A Jupyter Notebook or Colab notebook with complete code
2. At least two visuals such as class distribution or text length distribution
3. A short interpretation paragraph written in clear academic language

Submission Reminder

The notebook should be organized and readable. Use meaningful section titles such as dataset overview, data quality checks, class distribution, text characteristics, and interpretation.

Simple Scoring Guide

Use the checklist below to evaluate the submitted work for this activity.

Criteria	Description	Points

Correct Execution	Notebook runs and shows the required EDA steps.	10
Visual Outputs	At least two relevant visuals are included.	10
Interpretation	Observations are clear and connected to the dataset.	5

```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

file_path = 'data/stroke_prediction_dataset.csv'
df = pd.read_csv(file_path)
```

```python
print("\n5 Earliest Row")
print(df.head())
```

```
5 Earliest Row
   Patient ID       Patient Name  Age Gender  Hypertension  Heart Disease  \
0       18153    Mamooty Khurana   56   Male             0              1
1       62749  Kaira Subramaniam   80   Male             0              0
2       32145      Dhanush Balan   26   Male             1              1
3        6154        Ivana Baral   73   Male             0              0
4       48973  Darshit Jayaraman   51   Male             1              1

  Marital Status      Work Type Residence Type  Average Glucose Level  ...  \
0        Married  Self-employed          Rural                 130.91  ...
1         Single  Self-employed          Urban                 183.73  ...
2        Married   Never Worked          Rural                 189.00  ...
3        Married   Never Worked          Urban                 185.29  ...
4       Divorced  Self-employed          Urban                 177.34  ...

     Alcohol Intake Physical Activity Stroke History Family History of Stroke  \
0    Social Drinker          Moderate              0                      Yes
1             Never               Low              0                       No
2            Rarely              High              0                      Yes
3  Frequent Drinker          Moderate              0                       No
4            Rarely               Low              0                      Yes

   Dietary Habits Stress Levels Blood Pressure Levels  Cholesterol Levels  \
0           Vegan          3.48               140/108   HDL: 68, LDL: 133
1           Paleo          1.73                146/91    HDL: 63, LDL: 70
2           Paleo          7.31                154/97    HDL: 59, LDL: 95
3           Paleo          5.35                174/81   HDL: 70, LDL: 137
4      Pescatarian          6.84                121/95    HDL: 65, LDL: 68

                                                 Symptoms  Diagnosis
0                                Difficulty Speaking, Headache     Stroke
1                Loss of Balance, Headache, Dizziness, Confusion     Stroke
2                                           Seizures, Dizziness     Stroke
3  Seizures, Blurred Vision, Severe Fatigue, Headache, Confusion  No Stroke
4                                           Difficulty Speaking     Stroke
```

```python
print(f"\nDataset Size:")
print(f"Rows: {df.shape[0]}")
print(f"Columns: {df.shape[1]}")
```

```python
print("\nColumns and their Data Types:")
print(df.dtypes.to_string())
```

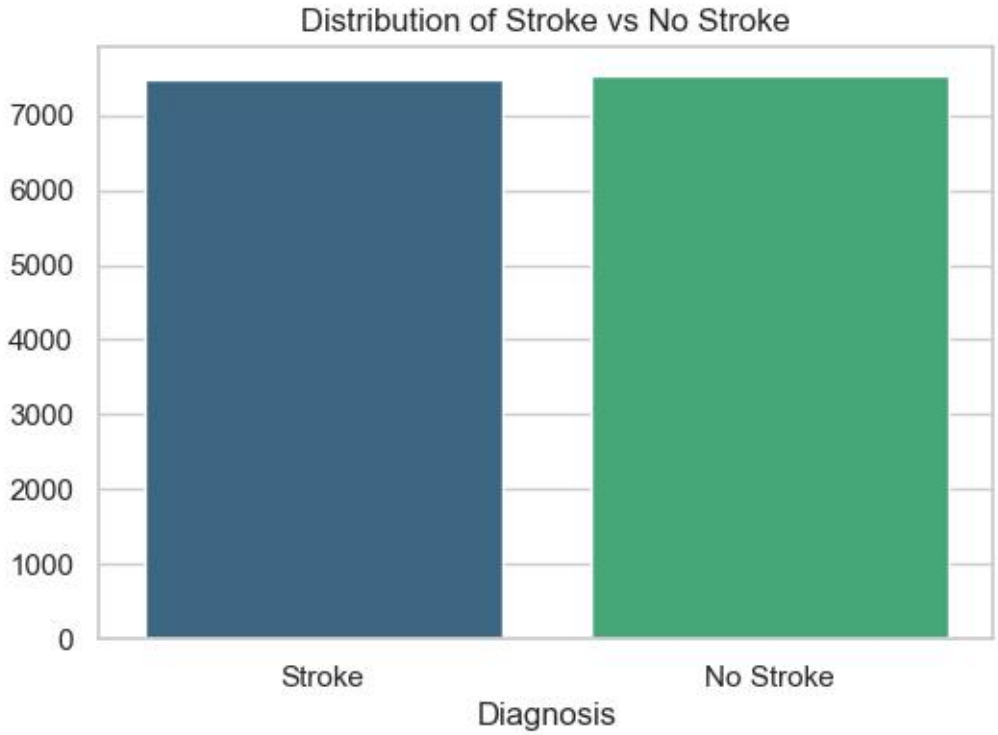
```
Dataset Size:
Rows: 15000
Columns: 22
```

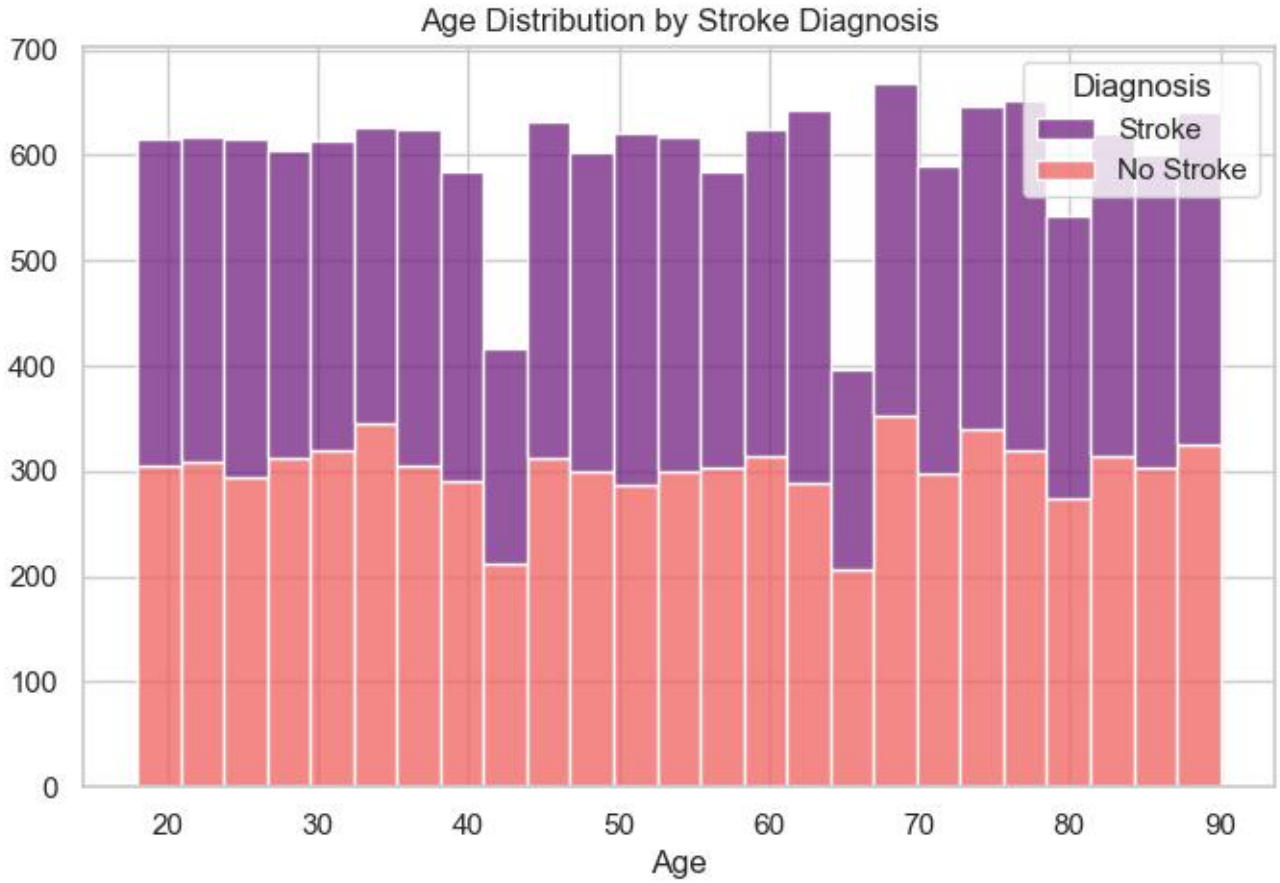
```
Columns and their Data Types:
Patient ID                    int64
Patient Name                    str
Age                           int64
Gender                          str
Hypertension                  int64
Heart Disease                 int64
Marital Status                  str
Work Type                       str
Residence Type                  str
Average Glucose Level       float64
Body Mass Index (BMI)       float64
Smoking Status                  str
Alcohol Intake                  str
Physical Activity               str
Stroke History                int64
Family History of Stroke        str
Dietary Habits                  str
Stress Levels               float64
Blood Pressure Levels           str
Cholesterol Levels              str
Symptoms                        str
Diagnosis                       str
```

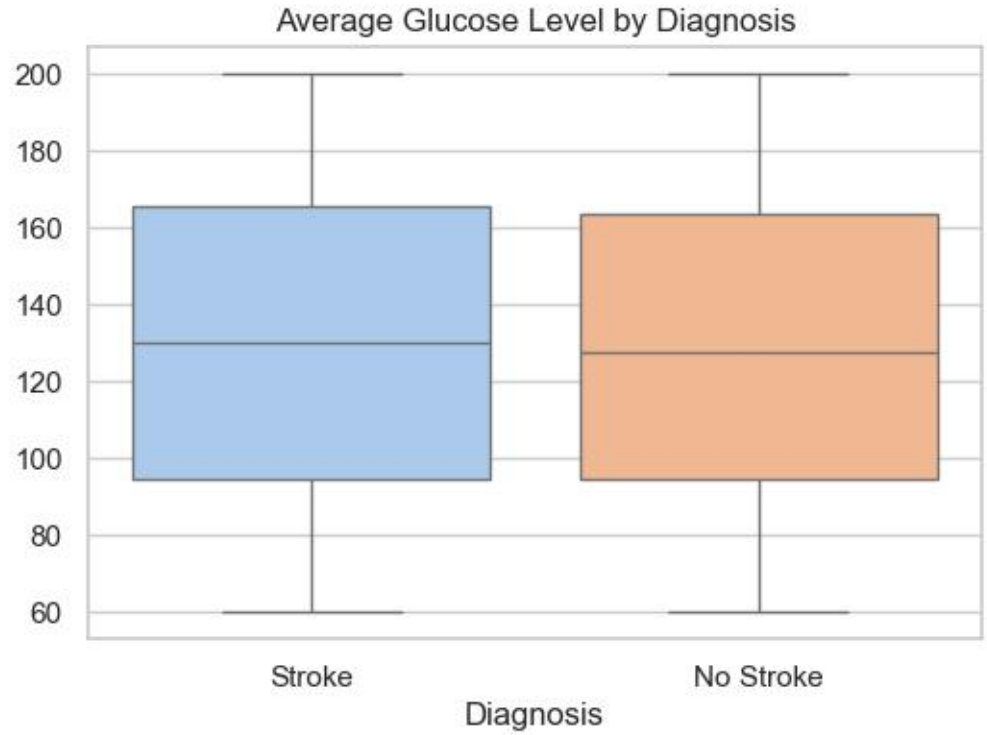
```python
print(f"\nData Quality Check:")
print(f"Missing Values:")
missing_data = df.isnull().sum()
print(missing_data[missing_data > 0].to_frame(name='Missing Count'))
if missing_data.sum() == 0:
    print("No missing values found!")
print(f"Duplicates found: {df.duplicated().sum()}")
```

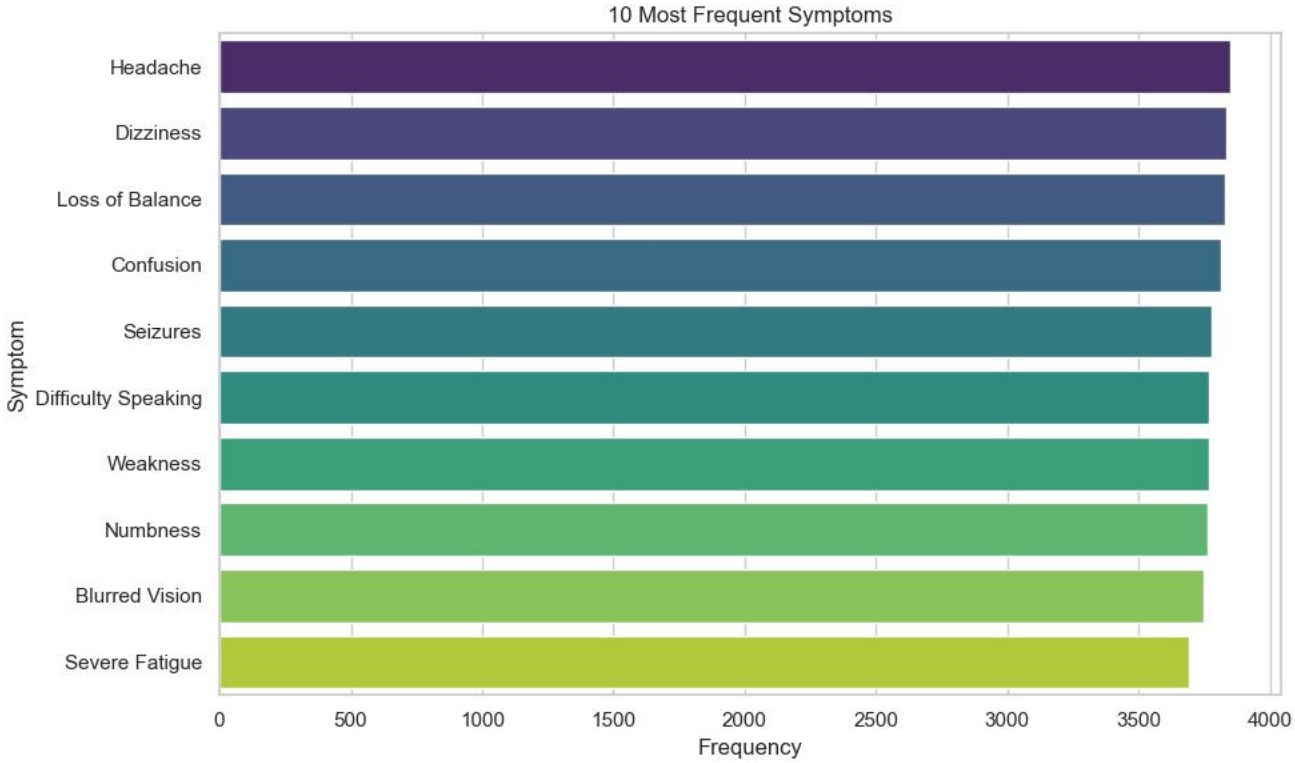
```
Data Quality Check:
Missing Values:
          Missing Count
Symptoms           2500
Duplicates found: 0
```

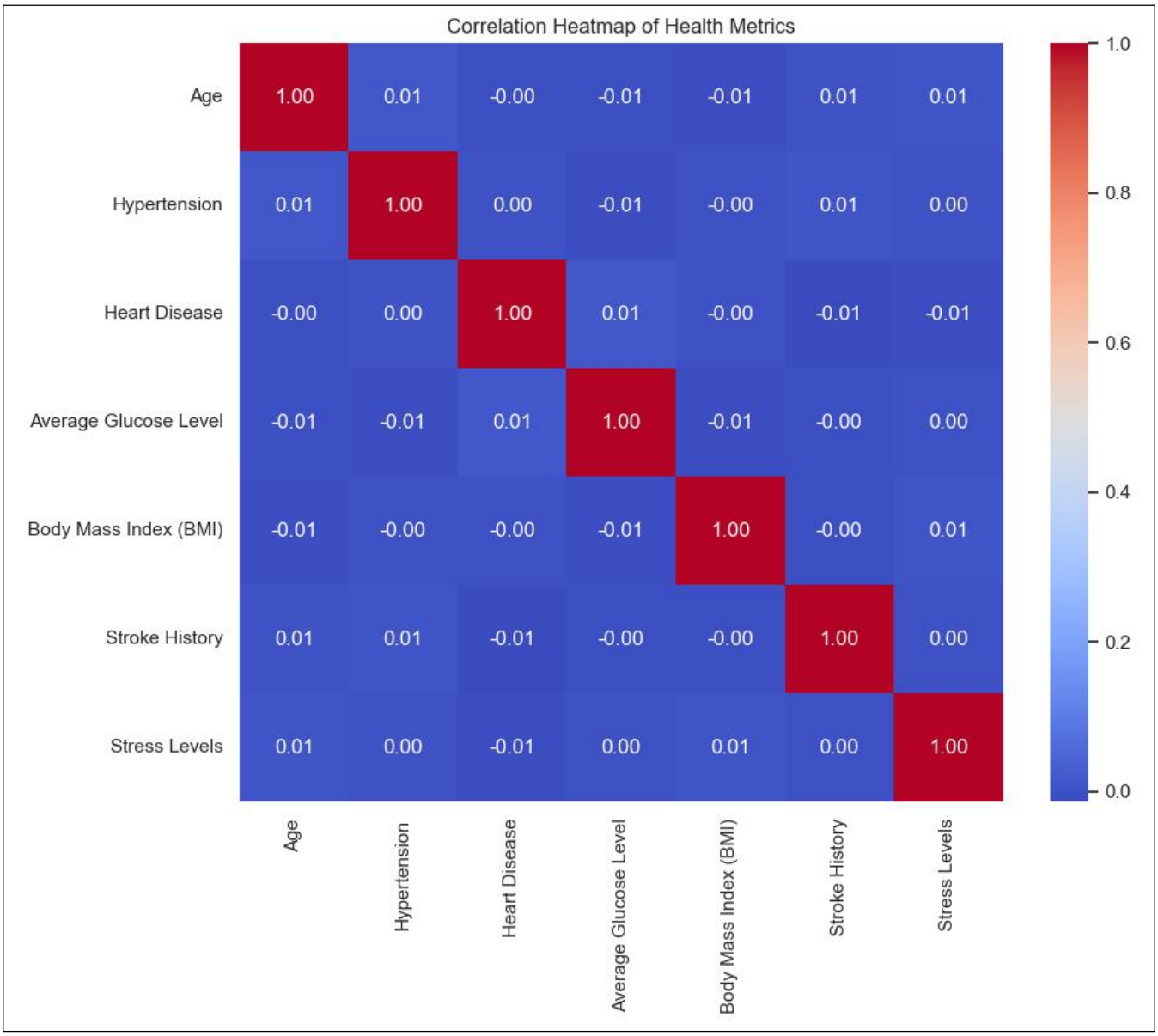
```
plt.figure(figsize=(6,4))
sns.countplot(data=df, x='Diagnosis', hue='Diagnosis', palette='viridis', legend=False)
plt.ylabel(None)
plt.title('Distribution of Stroke vs No Stroke')
plt.show()

plt.figure(figsize=(8,5))
sns.histplot(data=df, x='Age', hue='Diagnosis', multiple="stack", palette='magma')
plt.ylabel(None)
plt.title('Age Distribution by Stroke Diagnosis')
plt.show()

plt.figure(figsize=(6,4))
sns.boxplot(data=df, x='Diagnosis', y='Average Glucose Level', hue='Diagnosis', palette='pastel', legend=False)
plt.ylabel(None)
plt.title('Average Glucose Level by Diagnosis')
plt.show()

symptom_counts = (
    df['Symptoms']
    .dropna()
    .str.split(',')
    .explode()
    .str.strip()
    .value_counts()
    .head(15)
)

plt.figure(figsize=(10, 6))
sns.barplot(x=symptom_counts.values, y=symptom_counts.index, hue=symptom_counts.index, palette='viridis', legend=False)
plt.title('10 Most Frequent Symptoms')
plt.xlabel('Frequency')
plt.ylabel('Symptom')
plt.tight_layout()
plt.show()

plt.figure(figsize=(10,8))
numerical_df = df.select_dtypes(include=['float64', 'int64']).drop(columns=['Patient ID'])
sns.heatmap(numerical_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap of Health Metrics')
plt.show()
```


Distribution of Stroke vs No Stroke
7000
6000
5000
4000
3000
2000
1000
0
Stroke
No Stroke
Diagnosis


Age Distribution by Stroke Diagnosis
700
600
500
400
300
200
100
0
Diagnosis
Stroke
No Stroke
20
30
40
50
60
70
80
90
Age


Average Glucose Level by Diagnosis
200
180
160
140
120
100
80
60
Stroke
No Stroke
Diagnosis


10 Most Frequent Symptoms
Headache
Dizziness
Loss of Balance
Confusion
Seizures
Difficulty Speaking
Weakness
Numbness
Blurred Vision
Severe Fatigue
Symptom
0
500
1000
1500
2000
2500
3000
3500
4000
Frequency



The Stroke Prediction data analysis reveals a balanced dataset of 15,000 records. Key insights include a significant increase in stroke frequency among older populations (ages 60-90) and a high prevalence of neurological symptoms such as 'Difficulty Speaking' and 'Confusion.' .Metrics like Glucose and BMI showed overlapping distributions between classes, suggesting that predictive modeling will require a holistic analysis of multiple health factors rather than relying on a single clinical metric.